On Data: Collection & Creation

CS 485, Spring 2026
Applications of Natural Language Processing

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[Slides by Laure Thompson]

General NLP problem

Input: some text $\boldsymbol{x}$ (e.g. sentence, document)

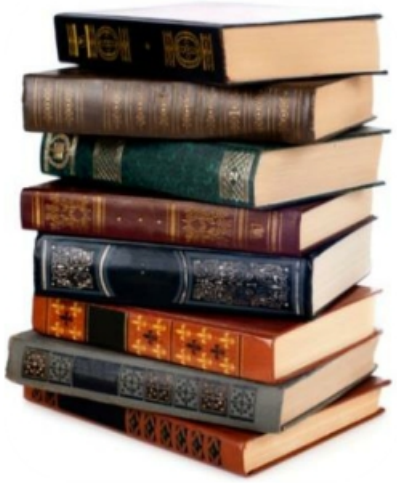
Output: a label $\boldsymbol{y}$ (from some finite label set)

Goal: learn a mapping function $\boldsymbol{f}$ from $\boldsymbol{x}$ to $\boldsymbol{y}$

What is a dataset?

A collection of texts and metadata

Text sources can be used in many ways



Text Unit

· Full Volume

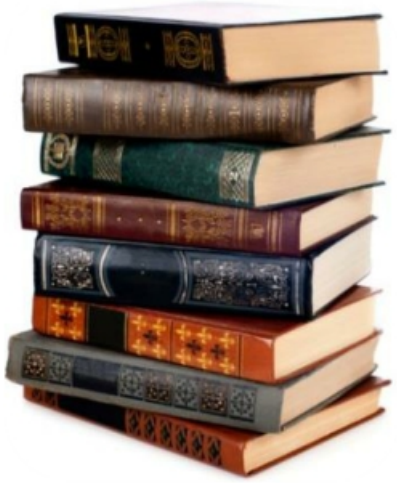
Metadata / Label Examples

· Author, Translator

· Genre, Literariness

· Publication Year

Text sources can be used in many ways



Text Unit

- Full Volume
- Passage

Metadata / Label Examples

- Author, Translator
- Genre, Literariness
- Publication Year
- Elapsed Narrative Time
- Event Depiction
- Sarcasm, Irony, Suspense
- Memorability

Text sources can be used in many ways

Text Unit

· Post

Metadata / Label Examples

· Popularity, Controversy

· Topic / Post Type

· Shared News Source

Text sources can be used in many ways

Text Unit

- Post
- Interaction (Post + Replies)

Metadata / Label Examples

- Popularity, Controversy
- Topic / Post Type
- Shared News Source
- Agreement / Disagreement
- Moderator Intervention
- Condescending Language Use

Where to get datasets?

Off the shelf ⟷ Build from scratch

Where to get datasets?

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Q: What are the pros & cons of each strategy?

Off - the - Shelf Datasets

There are many publicly available datasets.

- https://huggingface.co/datasets
- https://nlpprogress.com/
- Shared tasks: SemEval, CoNLL shared tasks, Kaggle (https://www.kaggle.com/datasets), …

Middle ground: Augmentation & Curation

Off the shelf

Build from scratch

Middle ground: Augmentation & Curation

Off the shelf ←——|——→ Build from scratch

- Use a subset of the texts based on metadata, content, etc.
- Combine texts from multiple data sources
- Align text and metadata from different sources
- Add new labels to an existing dataset

Example: CMU Movie Summary Corpus

Dataset: 42,306 movie plot summaries extracted from Wikipedia
+aligned metadata extracted from Freebase

http://www.cs.cmu.edu/~ark/personas/

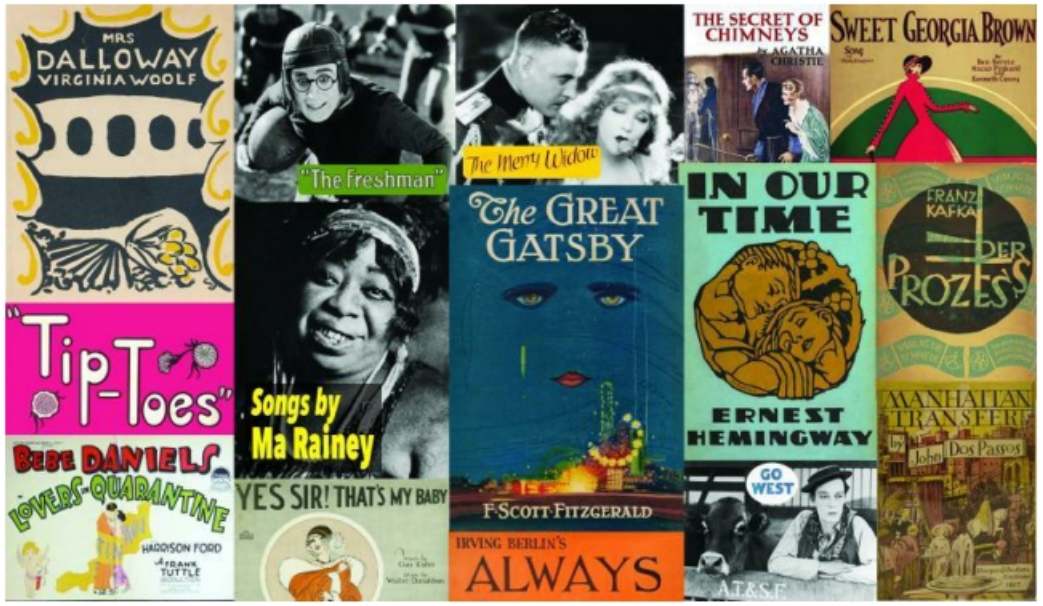
So you want to collect some data

1. Is it ethical to collect this data?
2. Is this data publicly available?
3. Is this data protected by copyright?

U.S. Public Domain

Published works enter the public domain **95 years** after their first publication date (in the U.S.)

As of this year, works published in **1925** entered the US public domain!



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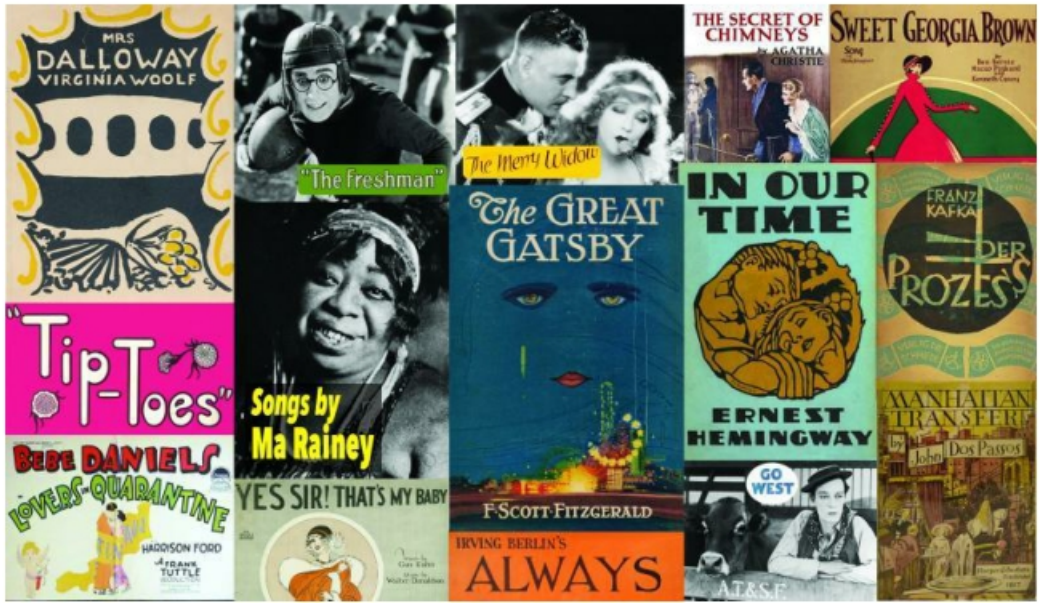

Books in the public domain: Project Gutenberg, https://www.gutenberg.org/

How to collect data?

- Digitization & Transcription
- APIs
- Bulk downloads / database dumps
- Web scraping

Digitization

Watch out for OCR errors!

THE WALWORTH MURDER.

The Story of a Member of the Bereaved Family.

Scenes Beside Chancellor Walworth's Coffin.

A Letter Written in Blood and Sanded with Powder.

A Wife's Silent Agony—Another Life To Be Saved at the Expense of the Memory of the Dead—Sad and Stern Facts which Must Come Out.

The walworth tragedy, which a few days ago startled the community to its depths, has not yet lost any of its morbid interest for the public. Every little detail and surrounding of the horrible murder, the antecedents of the victim and his slayer, even everything that remotedly concerns the bereaved family, is discussed with that unhealthy gout which only a satiety of crime can produce. The demeanor of the wretched prisoner, the characteristics of the father and the sufferings of the unfortunate mother are still subjects of all-absorbing interest in gossiping circles.

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APIs

Some examples:

· (now mostly closed) Twitter: twitter- api

· (now mostly closed?) Reddit: pushshift.io

Big problem: value of data and questions about tech company scrutiny

Q: What are the incentives for maintaining APIs?

Bulk Downloads

- Wikipedia: dumps.wikimedia.org/
- Caselaw Access Project: https://case.law/
- Pushshift still?? reddit.com/r/pushshift

Web Scraping Responsibly

Be considerate

- Check Terms of Service
- Check for robots.txt
- Use low request rates

Web Scraping Tools

- For fetching webpages from the command line: wget, curl
- … or, using Python: url lib

- For parsing .html files, for example to extract text content: BeautifulSoup, beautifulsoup4 python package

Curated balanced datasets

- What is “the English language in general” like? what different kinds of language are you exposed to during the day?

Curated balanced datasets

- Balanced corpora: Selected from different sources to mirror “the English language as a whole”
- Early example constructed for lexicographers: the Brown corpus
  - 1 million words
  - 500 samples of English, 2000+ words each, from works published in 1961
  - reportage, editorials, reviews, religion, skills/hobbies, popular lore, belles-lettres (memoirs and such), government & official, science&humanities, general fiction, mystery, sci fi, adventure, romance, humor

Curated balanced datasets

- More recent, larger corpora:
  - British National Corpus, 100M words
  - Corpus of Contemporary American English (COCA), 1.1B words
  - Corpus of Historical American English (COHA), 400M words

Books

Wikipedia derived datasets

Wikitext : High quality subset of Wikipedia pages.

WikiMatrix : Parallel sentences (across languages) of Wikipedia

SQUAD: Question - answer dataset that relies on Wikipedia text